

STELLAR EVOLUTION

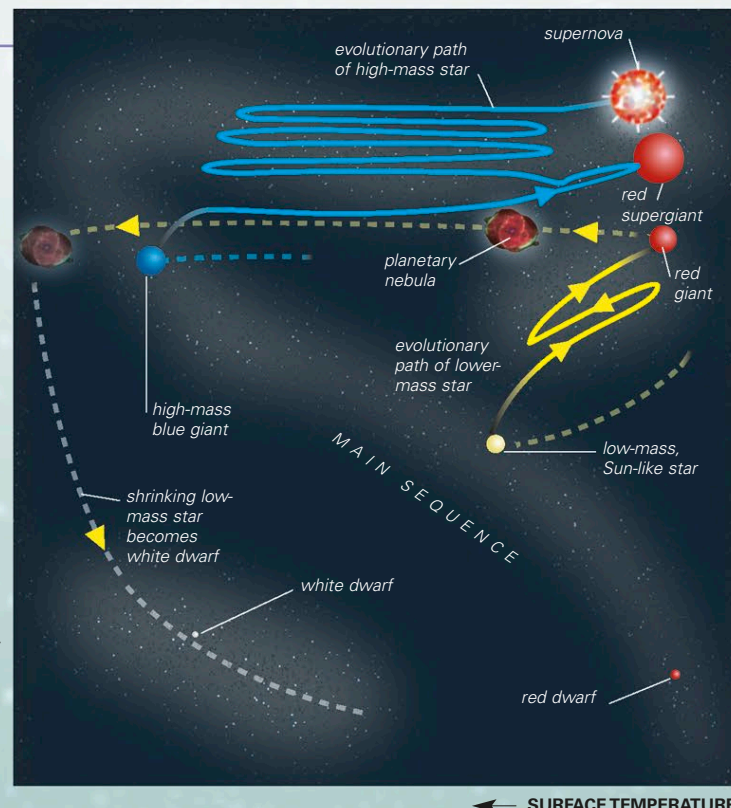
If they are of a sufficient mass, new stars will go onto the main sequence, where they will remain for most of their lives. When the hydrogen fuel in their cores is exhausted, they will evolve off the main sequence to become red giants or supergiants. Mass dictates what path stars will follow in their maturity. When a star expands as it burns fuel in its atmosphere or collapses after using up its fuel, it crosses a region to the right of the main sequence on the Hertzsprung–Russell (H–R) diagram (see p.232) known as an instability strip. The more massive the star, the more times it will expand and contract. High-mass stars explode as supernovae in the supergiant region of the H–R diagram, while low-mass stars cross back over the main-sequence band as they collapse to form white dwarfs. Being small and hot, white dwarfs appear in the bottom left of the H–R diagram. As they cool, they move to the right, eventually cooling to become black dwarfs. Neutron stars and black

holes do not appear on the H–R diagram because they do not fit the mass–luminosity relationship that it represents.

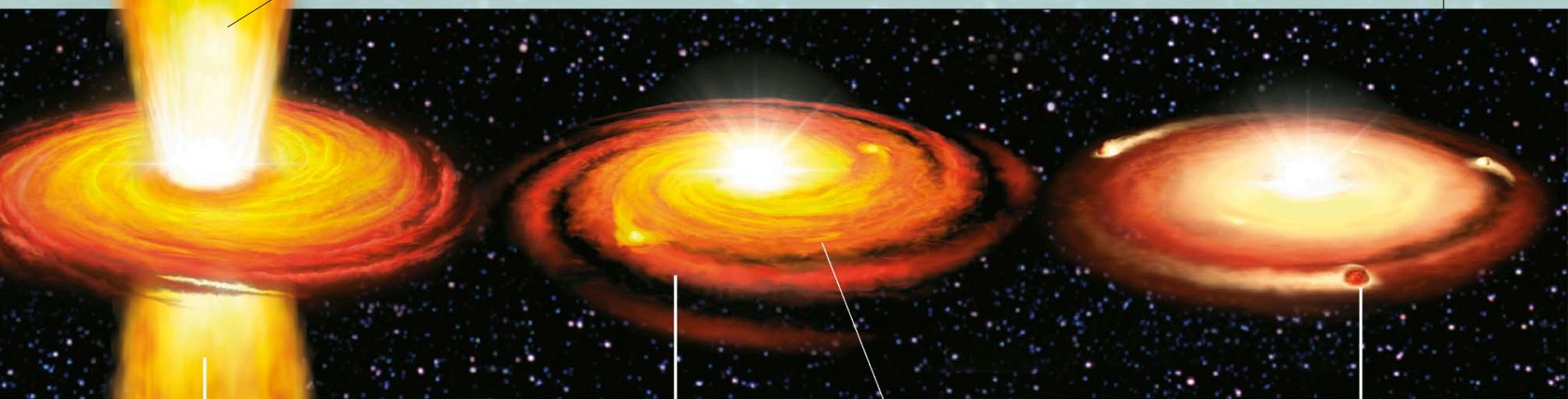
STELLAR MATURITY

The paths of mature stars on their journey toward death can be traced on the H–R diagram. Stars expand off to the right as they get larger and cooler to become red giants or supergiants. They travel back leftward as they collapse, after burning fuel in their atmospheres.

LUMINOSITY ↑



← SURFACE TEMPERATURE



STELLAR ADOLESCENCE

The gas that contracts to make a protostar starts to rotate slowly and speeds up as it is pulled inward, creating a disk of stellar material. Before joining the main sequence, the protostar exhibits unstable behavior such as rapid rotation and strong winds.

STAR REACHES MAIN SEQUENCE

For protostars with a mass of more than about 0.08 solar masses, the pressure and temperature within become high enough for nuclear reactions to start. The pressure balances gravity, and the protostar becomes a star.

circumstellar disk

FORMATION OF ORBITING PLANETS

Once a star is on the main sequence and stable, any disk of remaining material will start to cool. As it cools, elements condense out and begin to stick together. The larger clumps attract the smaller ones until conglomerations are planet-sized.

FORMATION OF A PLANETARY SYSTEM

Most young stars, unless they are in a close binary system, are surrounded by the remnants of the material from which they have formed. Rotation and stellar winds often shape the material into a flattened disk around the equatorial radius. Initially, the disk of material is hot, but as the star settles down onto the main sequence, it begins to cool. As it cools, different elements condense out, depending on the disk's temperature. Elements can exist in different states throughout the disk. Moving out from the star, temperatures fall, so water, for example, will exist as ice far away from the star and steam close to the star. Tiny condensing particles gradually stick together and grow larger. The ones that grow fastest will gravitationally attract others, becoming larger still, though in the dynamic early stages they may be broken back into pieces by collisions with other growing particles. Eventually, as the disk cools down, it becomes a calmer environment, and some particles will grow large enough to be classed as planetesimals—embryonic planets. Remnants of the original disk that do not form planets become asteroids or comets, depending on their distance from the parent star. Atmospheres are formed by gas attracted from the circumstellar disk, from gases erupting from the planets, or from bombarding comets.

CIRCUMSTELLAR DISK

Within the circumstellar disk of AB Aurigae, knots of material may be in the early stages of planet formation. This swirling disk of stellar material is about 30 times the size of the solar system.

